

MATH 110C, HOMEWORK 2

DUE TUESDAY APRIL 23

Exercise 1 Let F be a finite field and n a positive integer. Let K and K' be two finite extensions of F of degree n . Show that K and K' are F -isomorphic.

Exercise 2 Let p be a prime, n a positive integer and K a finite field with p^n elements. Let d be an integer with $1 \leq d \leq n$ and $d|n$. Show that K has a unique subfield with p^d elements.

Exercise 3 Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be a morphism of fields. Show that σ is the identity (show first that σ sends positive numbers to positive numbers).

Exercise 4 Determine the splitting field of $X^4 - 9$ in $\mathbb{Q}[X]$.

Exercise 5 Do Problems 1(e), 2(b), 5, 6 and 7 in Exercises 50.18, pp.307–308 of the book.